

Jaloliddin Ismailov

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EDUCATION

Eötvös Loránd University (ELTE)

BSc in Computer Science (GPA: 4.6/5.0; Member of Central Asian Society)

Budapest, HU

Expected: Jun 2028

EXPERIENCE

BME Solar Boat Team

Firmware & Embedded Systems Engineer

Budapest, HU

Feb 2026 – Present

- Developing safety-critical embedded firmware using C and FreeRTOS on ESP32, optimizing low-level peripheral drivers (I2C/SPI) to guarantee reliable throttle operation and real-time system stability.
- Architected robust data acquisition and sensor fusion pipelines featuring bit-level error detection and scaled ADC telemetry, enabling real-time edge processing and Hardware-in-the-Loop (HIL) testing to mitigate critical failures.

CERN

Software Engineering Research Fellow (LHCb)

Remote / Geneva, CH

Nov 2025 – Jan 2026

- Engineered high-throughput muon track reconstruction algorithms using Object-Oriented C++ and parallelized GPU computing (CUDA), optimizing model inference and large-scale distributed data pipelines for particle physics.
- Streamlined Agile software development lifecycles by automating dependency management, Docker containerisation, and CI/CD pipelines, significantly accelerating backend deployment workflows.

PROJECTS

Assistive Smart Glasses

Hardware & Edge AI Developer [Video Demo]

BME Robotics Hackathon

Jun 2026

- Engineered an edge AI-capable system architecture using C++ and PlatformIO, deploying TinyML concepts and computer vision algorithms on an ESP32-CAM to process and transmit real-time spatial telemetry for active navigation.

Flood-Risk Monitoring System

Backend Developer [Video Demo]

CASSINI Hackathon Budapest

Apr 2026

- Built a scalable geographic mapping backend utilizing Python, GeoPandas, and RESTful APIs, deploying data analytics models to transform raw environmental sensor data into actionable spatial visualisations.

EXTRACURRICULARS

European Young Engineers

Regional Coordinator

Remote

Mar 2026 – Present

- Analyzing the intersection of applied machine learning integration and regional climate policies across Hungary, Austria, and Slovakia to facilitate cross-border strategic planning.

FLEX Program (U.S. Department of State)

Exchange Student

Minnesota, USA

Sep 2023 – Mar 2024

- Completed 50+ hours of community service and collaborated with the International Student's Club to foster cross-cultural communication and relationship-building frameworks.